Refractive Errors - Year IV– 11/3/2016

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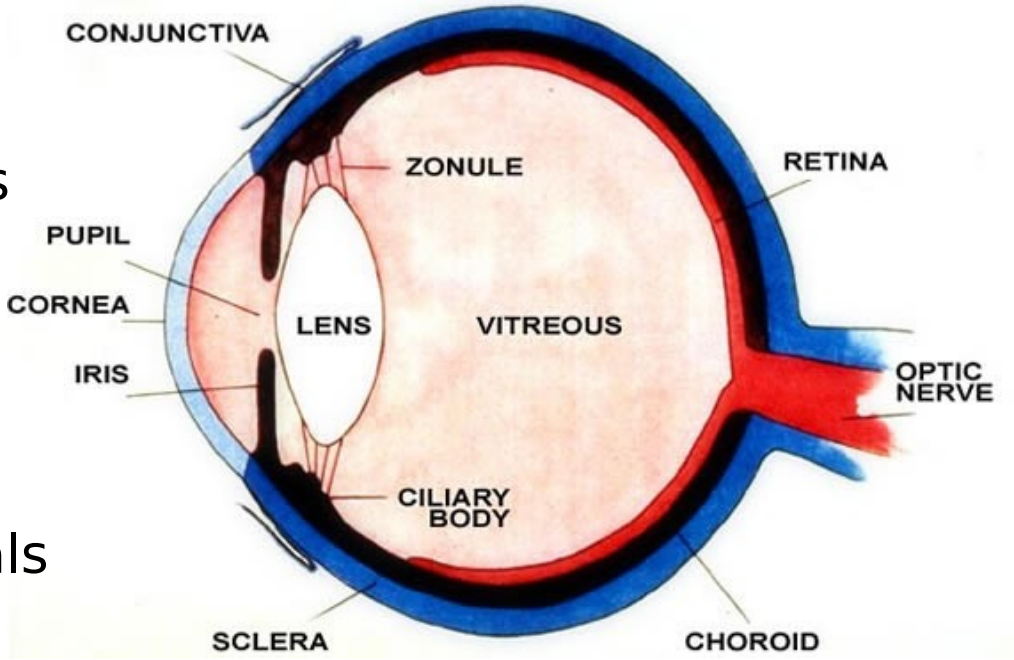
Objectives

- Understand normal vision
- Definitions
  - Myopia
  - Hyperopia
  - Astigmatism
  - Presbyopia
- Detection
- Treatment
  - Spectacle correction
  - Contact lenses
  - LASIK

Normal Vision

Light eye thro’

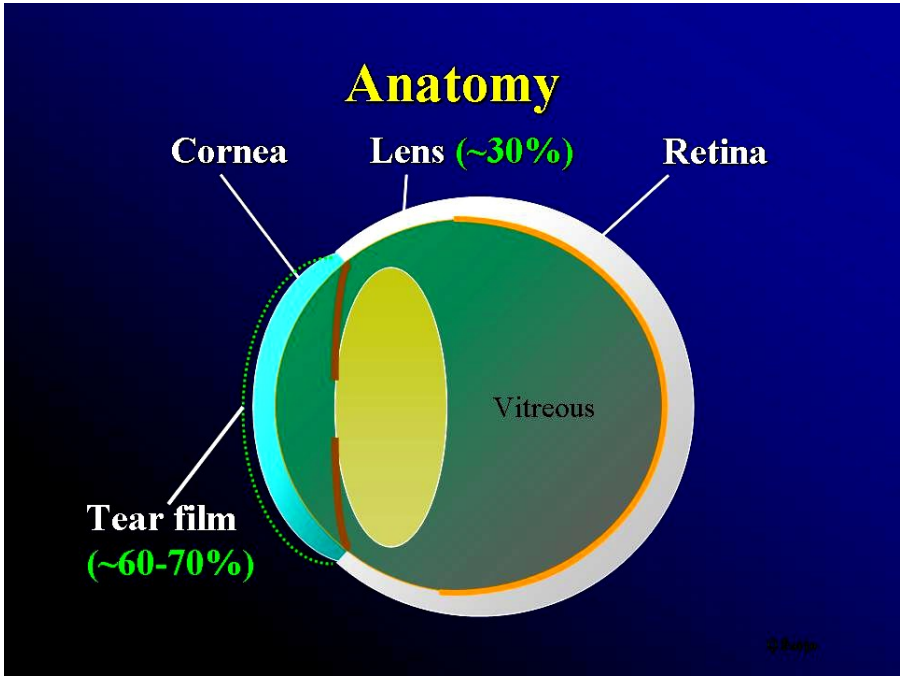
- eye cornea
- pupil – regulation of light
- hits the lens
- Passes through the vitreous humor
- retina - image appears in-verted
- ON carries signals to visual cortex - assembles the signals into images



Definitions

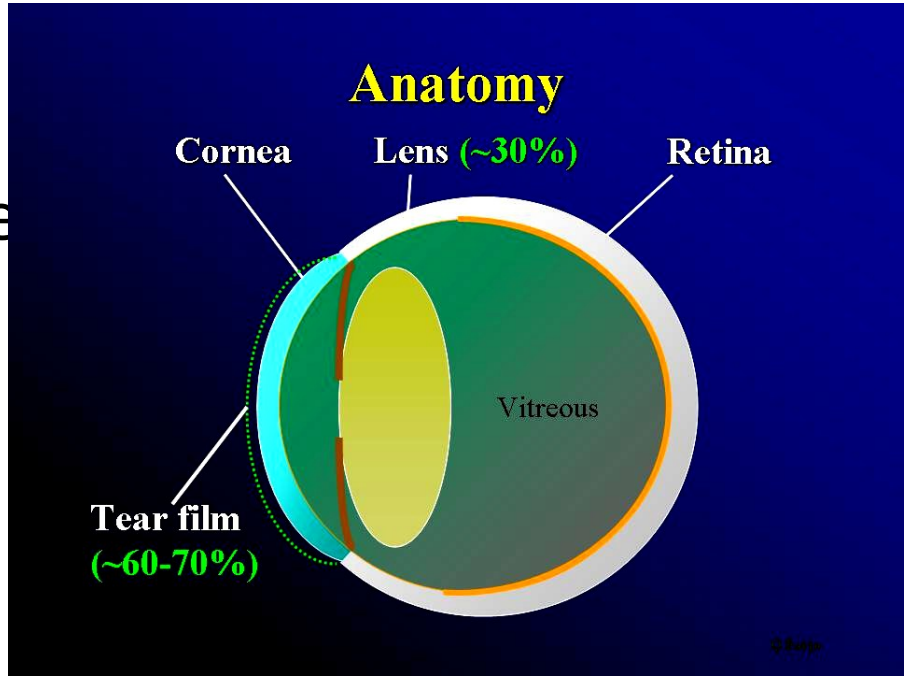
Refraction

- Ratio of the refractive power of the *lens* and *cornea* (**refractive media**) to the axial length of the globe
- Refractive power (**RP**) is specified in diopters **- D**



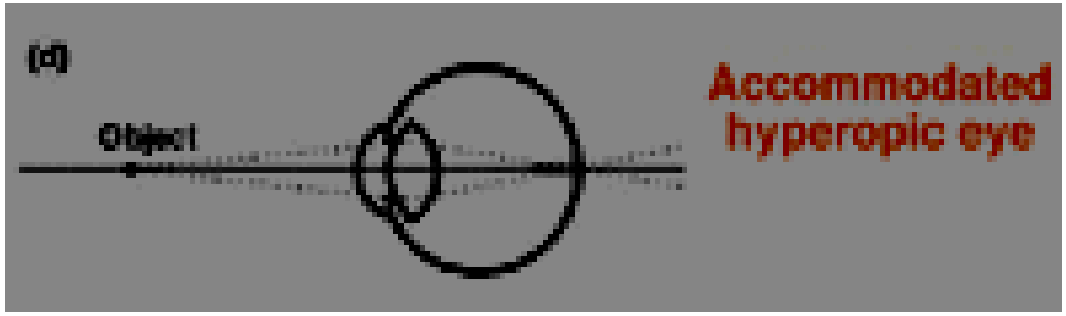
Definitions…..

- RP is calculated according to laws of geometric optics
- Maximum total refractive power - emmetropic eye is 63D with an axial length of 23.5
  - (Cornea- 43D; lens – 20D)



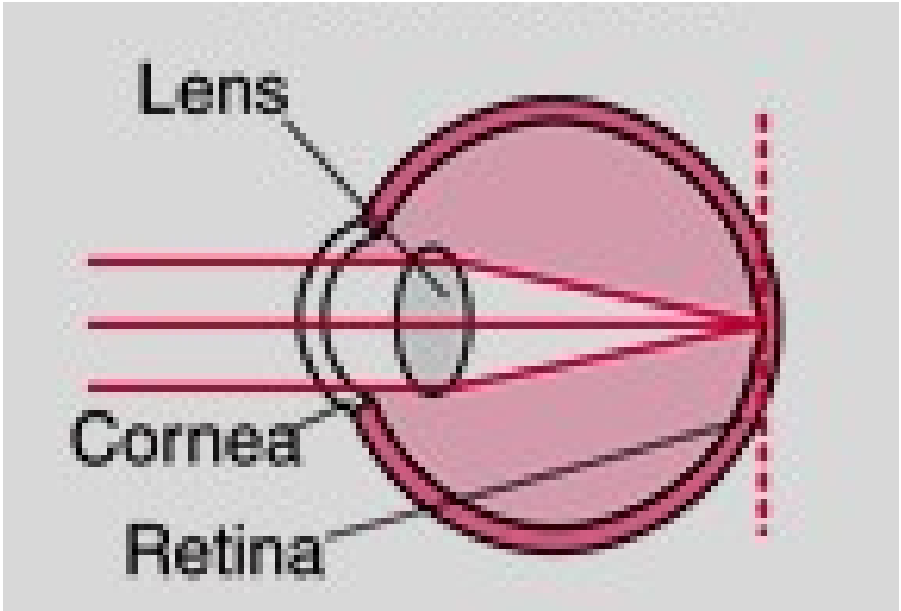
Definition……

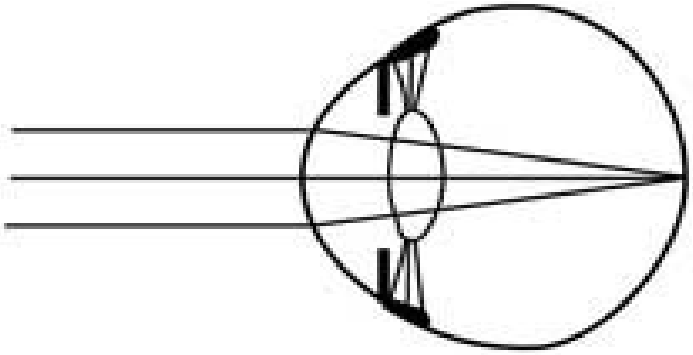
- Accommodation
  - The RP of the eye is not con-stant – alters for near and far
  - Involves
    - The lens,
    - Zonule fibres
    - Ciliary muscle



Emmetropia (normal sight)

- The ratio of refractive media:axial length is balanced
- Parallel rays meet at a focal point on the retina





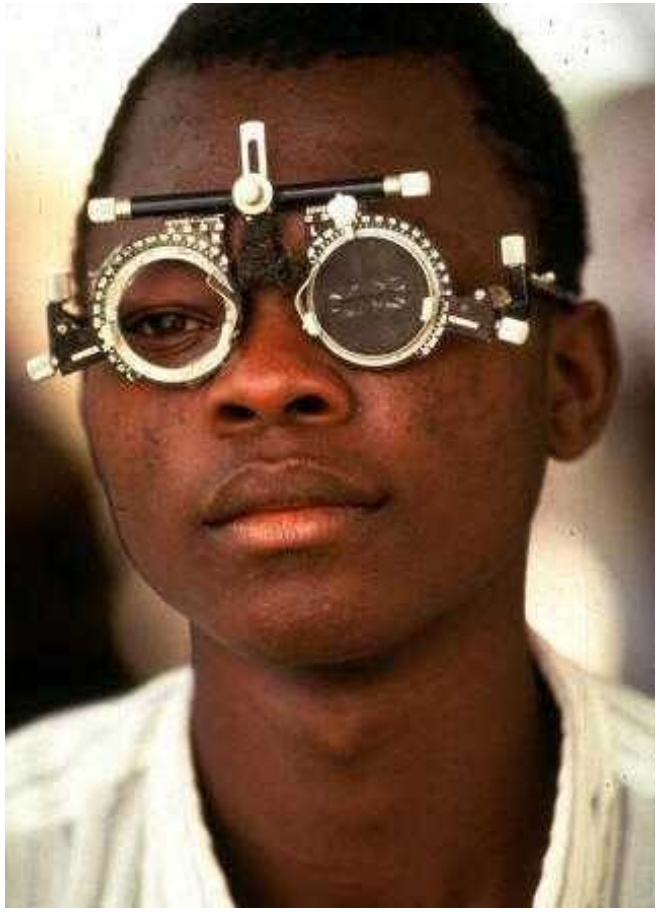
Normal eye

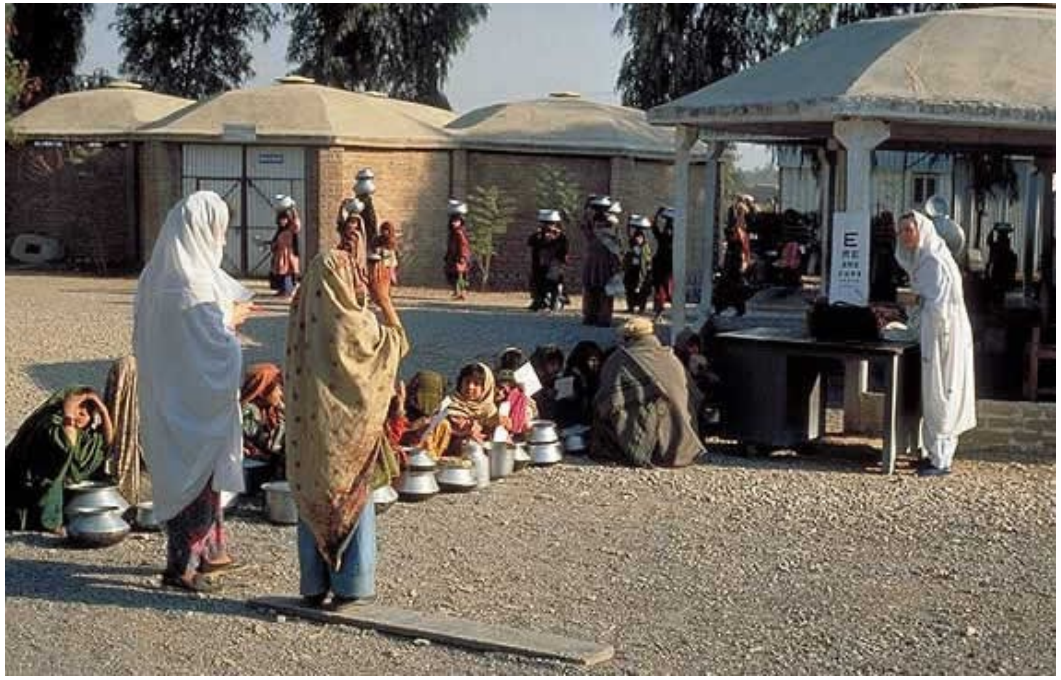
Ametropia (refractive err

Definition:

- Mismatch between axial length of the eye and the RP of the cornea and the lens
  - Axial (common)
    - Myopia
    - Hypermetropia: (Presbyopia)
    - Astigmatism
  - Refractive
    - Cornea
    - Lenticular

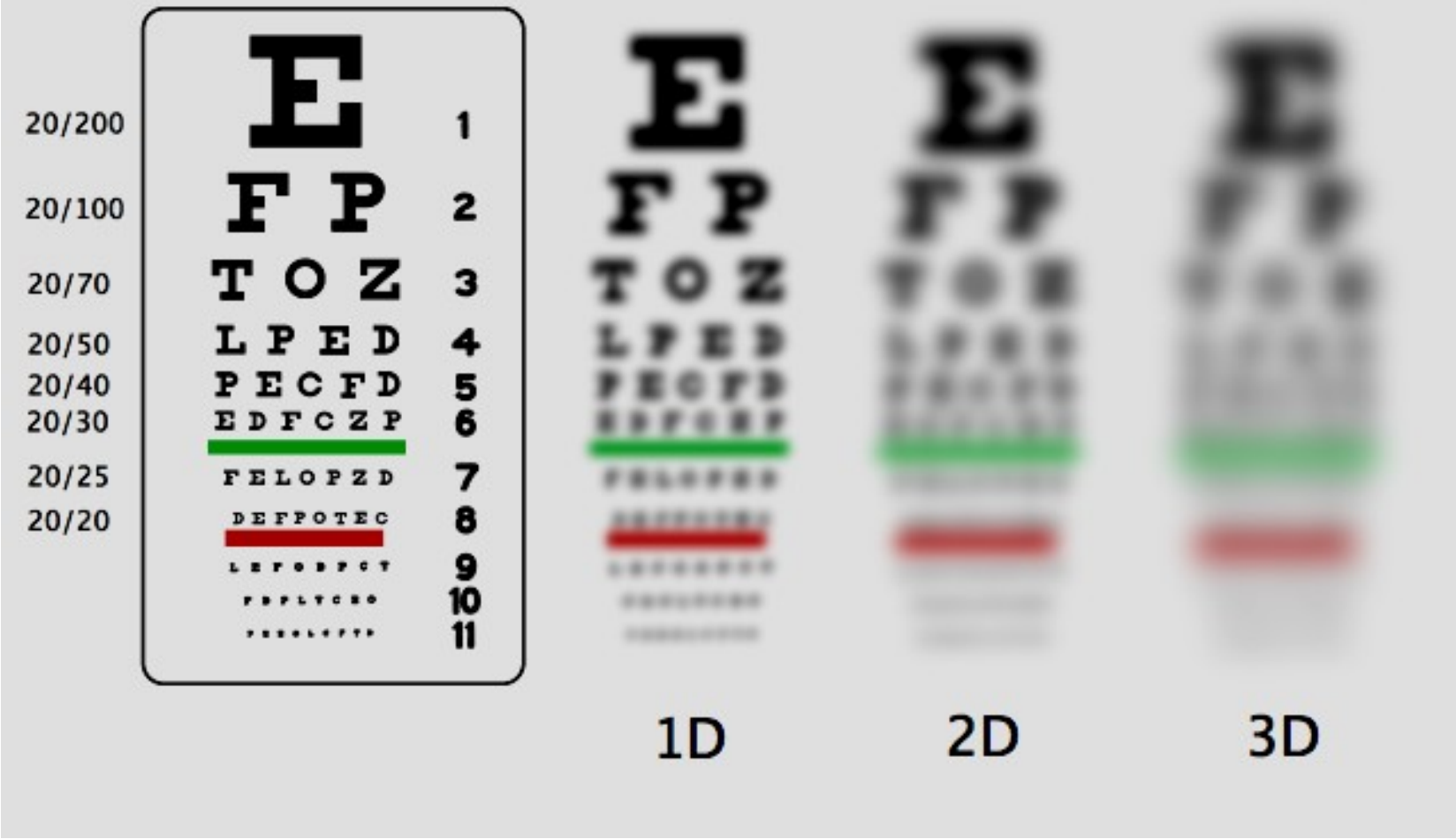
Testing Visual Acuity





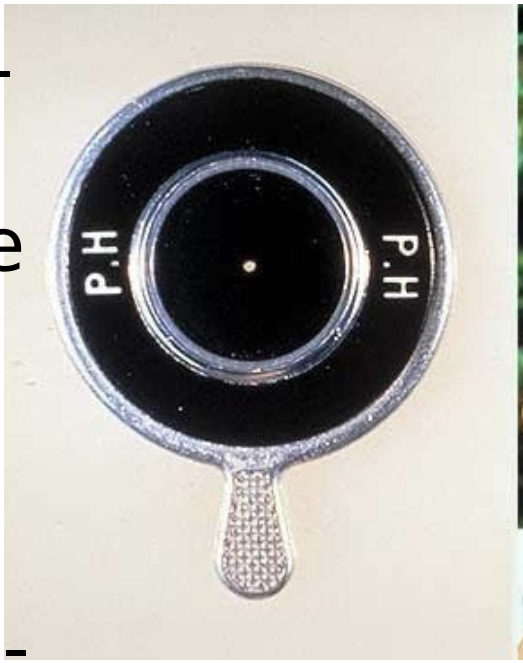
Examination

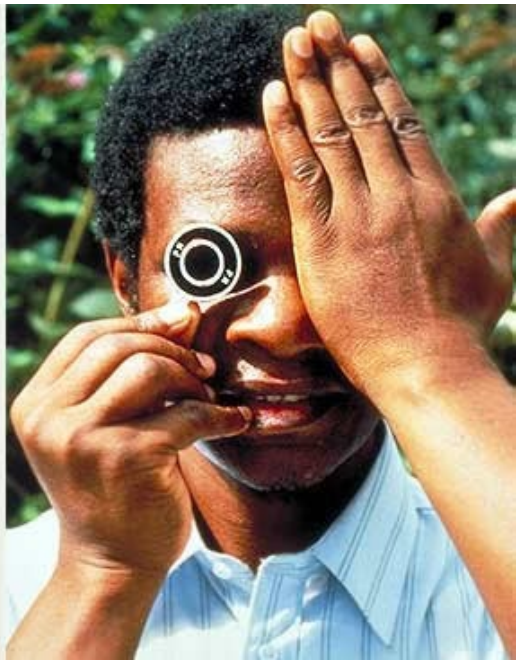
- Visual Acuity



The Pin Hole Test

- Testing Distance Vision with the Pin-Hole
- Pinhole - allows the passage of light which is perpen-dicular to the lens; does not need to be bent prior to be-ing focused onto the retina



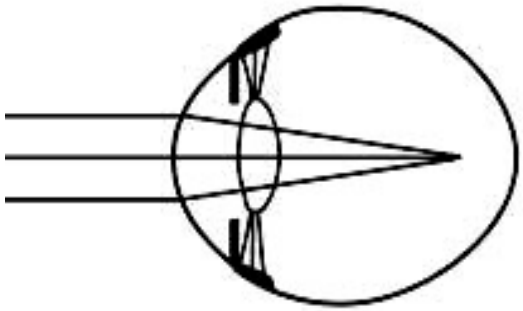


Snellen Notation Metric	Imperial	MAR	Log MAR	Decimal
6/60	20/200	10	1.0	0.10
6/48	20/160	8.0	0.9	0.13
6/38	20/125	6.3	0.8	0.16
6/30	20/100	5.0	0.7	0.20
6/24	20/80	4.0	0.6	0.25
6/19	20/60	3.2	0.5	0.32
6/15	20/50	2.5	0.4	0.40
6/12	20/40	2.0	0.3	0.50
6/9.5	20/30	1.6	0.2	0.63
6/7.5	20/25	1.25	0.1	0.80
6/6	20/20	1.00	0.0	1.00
6/4.8	20/16	0.80	-0.1	1.25
6/3.8	20/12.5	0.63	-0.2	1.58
6/3.0	20/10	0.50	-0.3	2.00

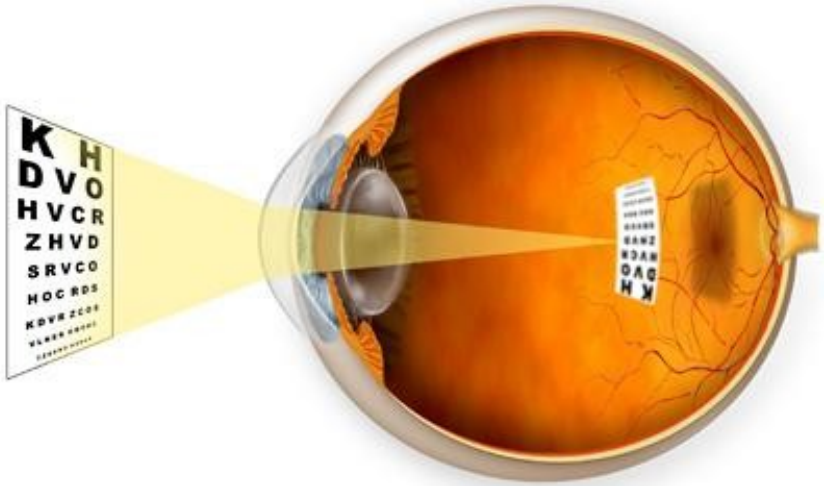
Near Vision Testing

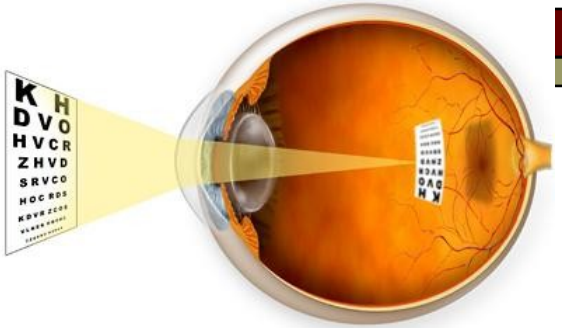
Myopia

- Discrepancy between the RP and the axial length – parallel rays converge at a focal point anterior to the retina



Light focused in front of retina

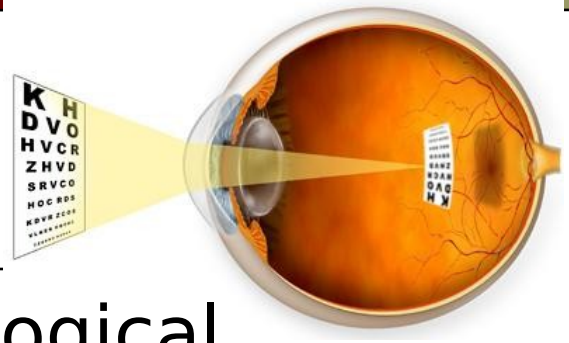




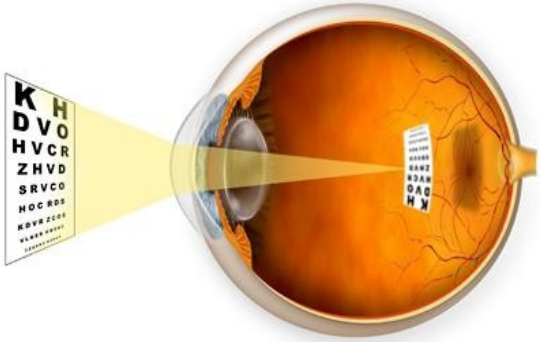
Myopia……

- Epidemiology
  - Generally: 25% of persons between 20 – 30 yrs (-1.0D)
  - Africa:
    - Nairobi (Kenya) - 10% 13-15yrs
    - Makueni (Kenya) – 2% 12-15 yrs

Myopia……

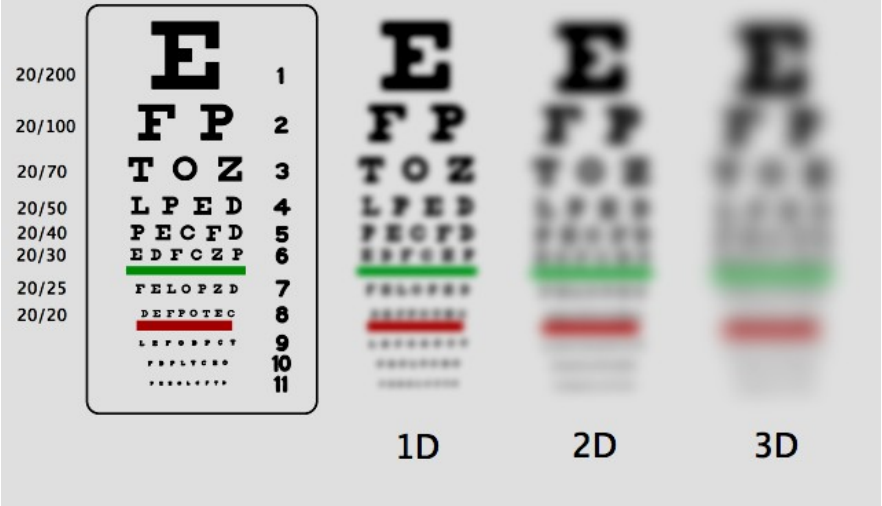


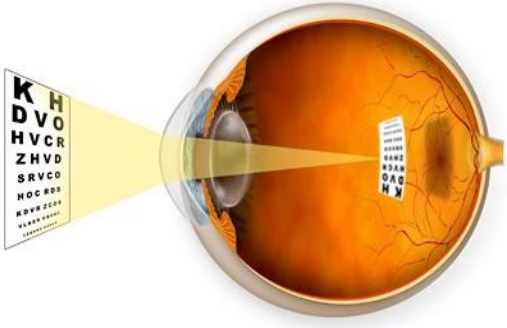
- Simple
  - Onset is 10 -12 years
  - Does not progress > 20 years (may have benign progressive – stabilizes at 30 yrs)
  - Refraction rarely > 6 Diopter
- Pathological
  - Largely hereditary
  - Varies in ethnic groups
  - Continuous progression
- Special forms of My-opia:
  - Lens-
    - Nuclear sclerosis
    - Spherophekia
  - Cornea
    - Keratoconus



Myopia……

- Symptoms
  - Inability to see distant objects
  - “Squinting” to focus
  - Can read -without glasses (older myopes)
  - Simulation of esotropoia (shift in axes of the eye)
  - Floaters: Reduction in volume of vitreous, collapse, vitreous opacification, Increased risk of RD





Myopia……

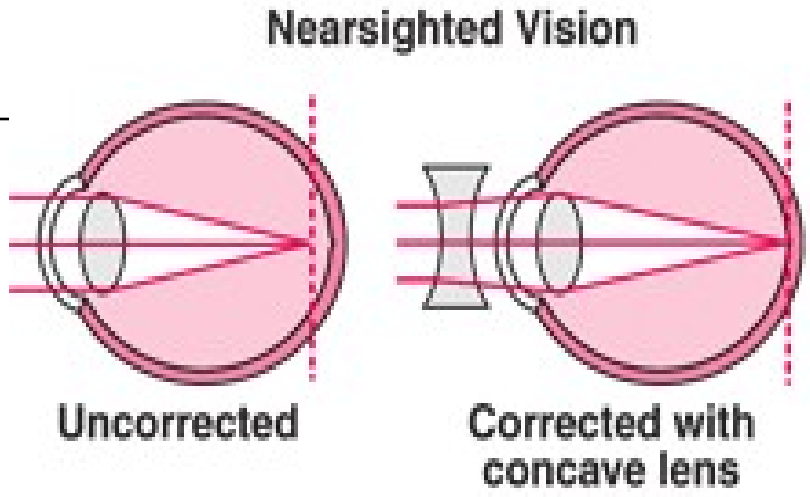
- Signs
  - Thinning of sclera
  - Deep anterior chamber
  - Atrophy of the ciliary muscle
  - Chorioretinal atrophy around O
  - Breaks in bruchs membrane
  - Subretinal neo-vascularization – retinal oedema/bleeding (Fuchs black sport)
  - Glaucoma
  - Optic nerve - oblique

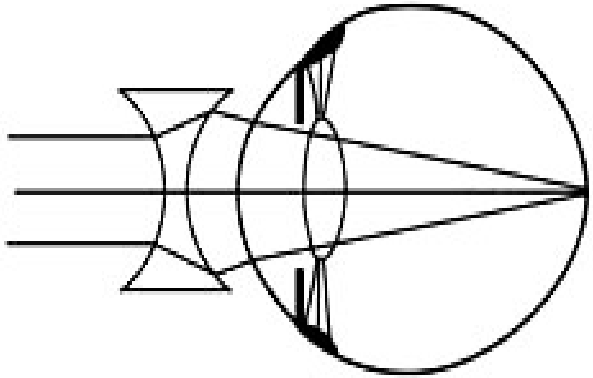
- Diagnosis
  - Refraction
    - Objective (Cy-cloplegic)
    - Subjective

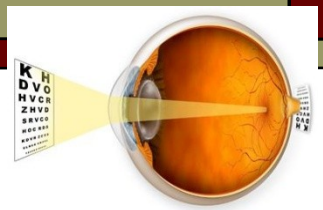
Myopia……

Treatment:

- Concave lenses
- Contact lenses
- Surgery
  - Removal of lens
  - LASIK

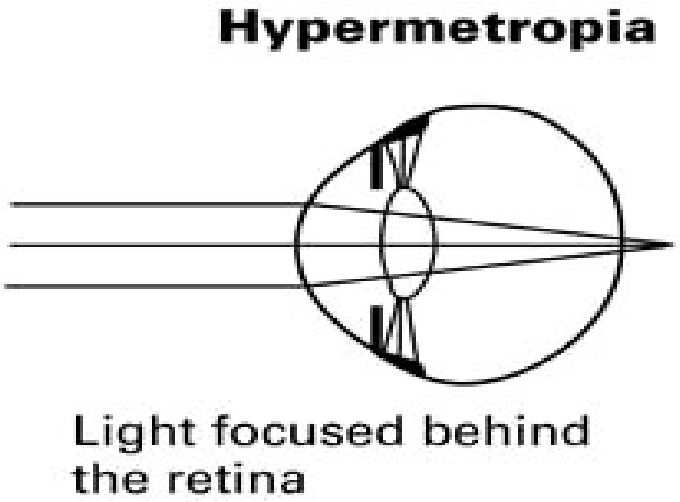


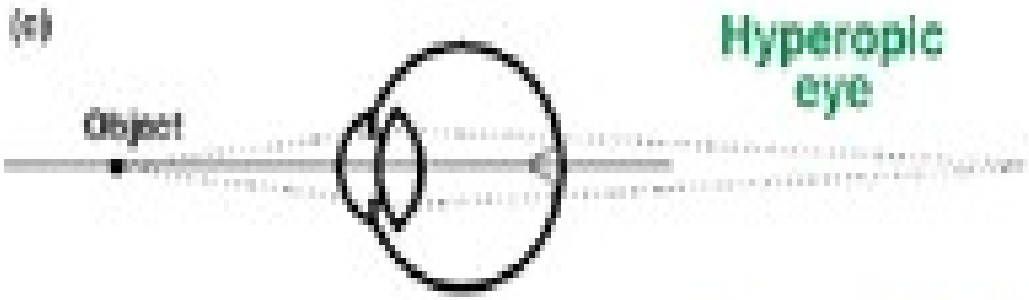


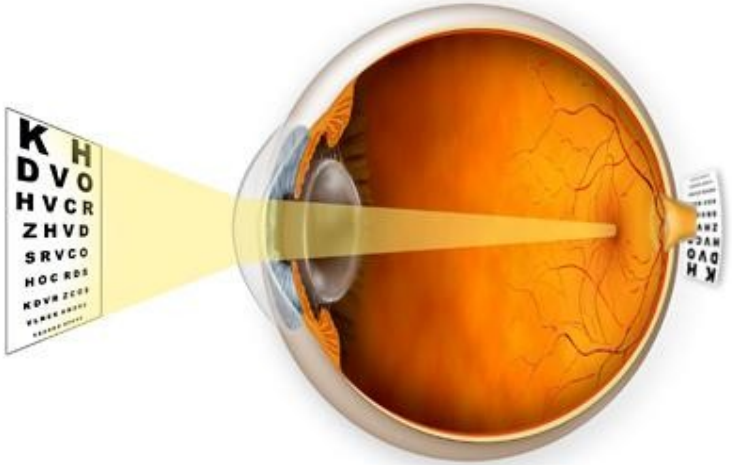


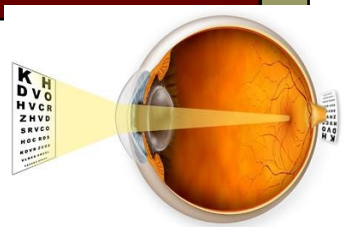
Hypermetropia…

- Definition
  - Discrepancy between the refractive power and the axial length – parallel rays converge at a focal point posterior to the retina



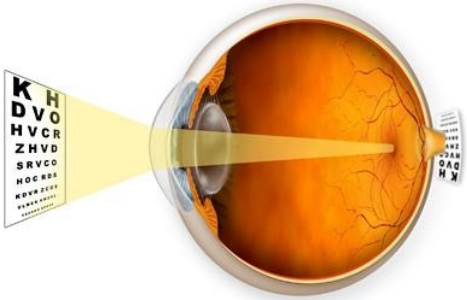






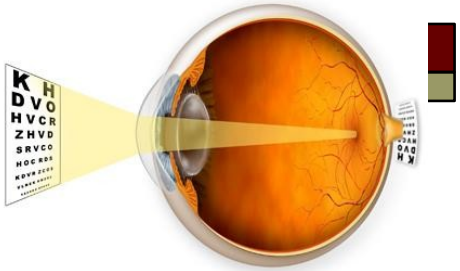
Hypermetropia.....

- Epidemiology
  - About 20% of ages 20-30yrs >+1.00DS
  - Most newborns
- Etiology
  - Exact mechanism not known
  - Special forms
    - Apakia: Congenital; Traumatic; Post Cataract Surgery



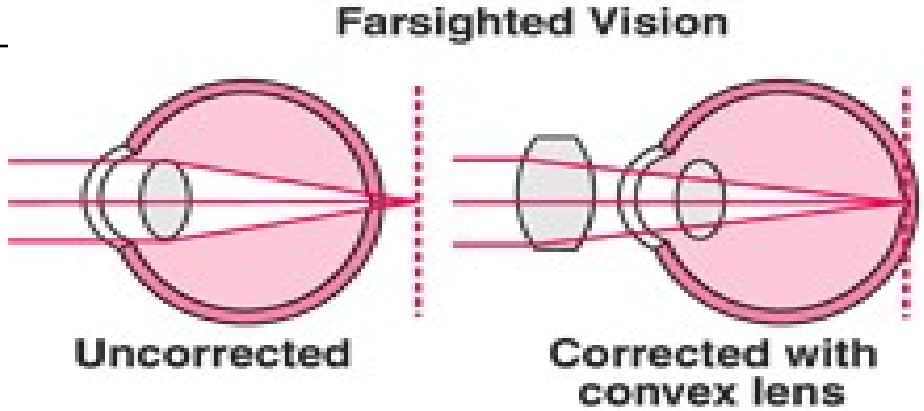
Hypermetropia……

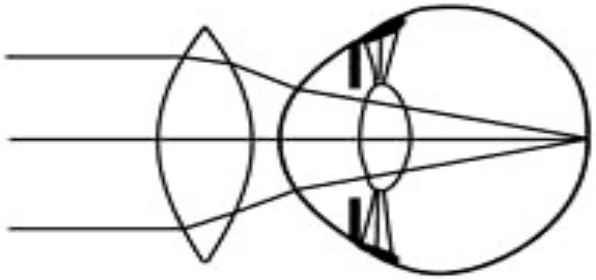
- Symptoms
  - asthenopic symp-toms: eye pain; headache; Blurred vision; Fatigue
  - Esotropia
  - Early presbyopia
- Signs
  - Blurred disc
  - Tortuous retinal ves-sels
  - Latent hyperopia
  - refraction



Hypermetropia

- Treatment
  - Concave specta-cles
  - Contact lenses
  - Surgery
    - LASIK

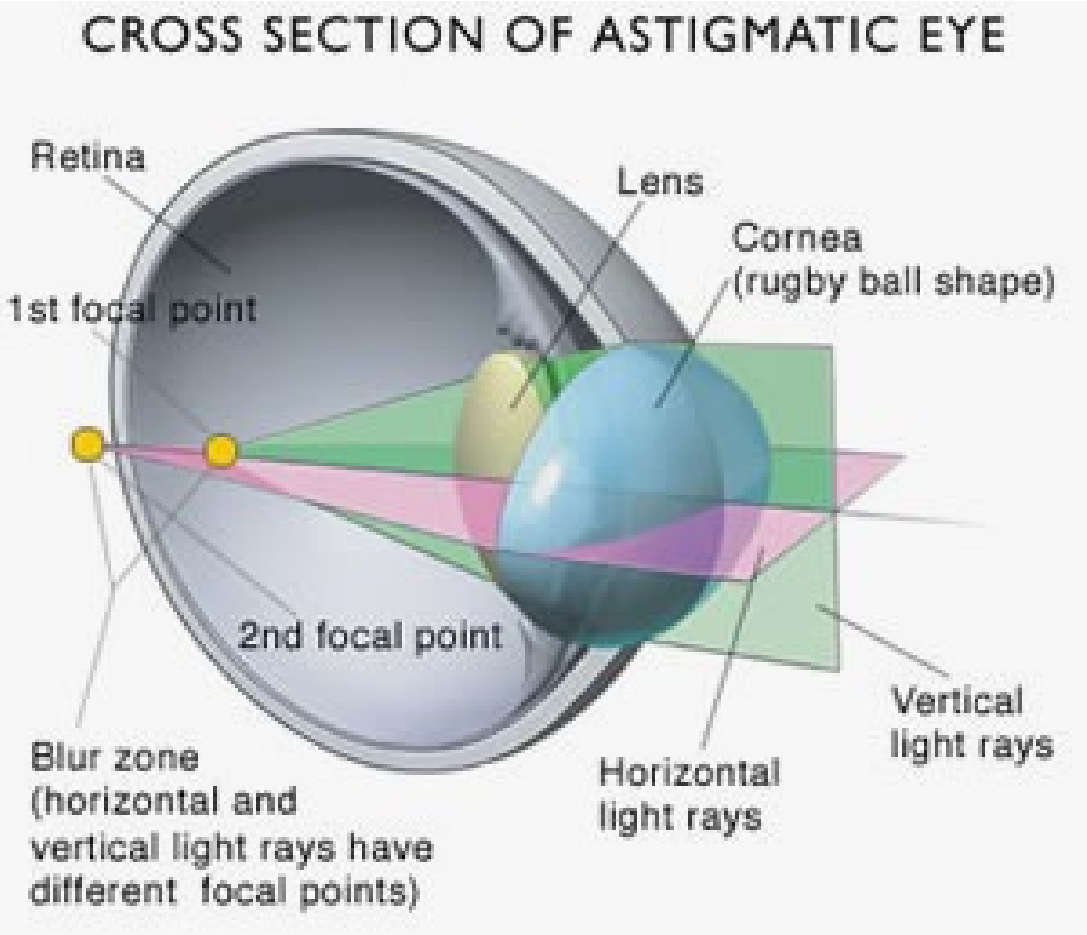




Astigmatism

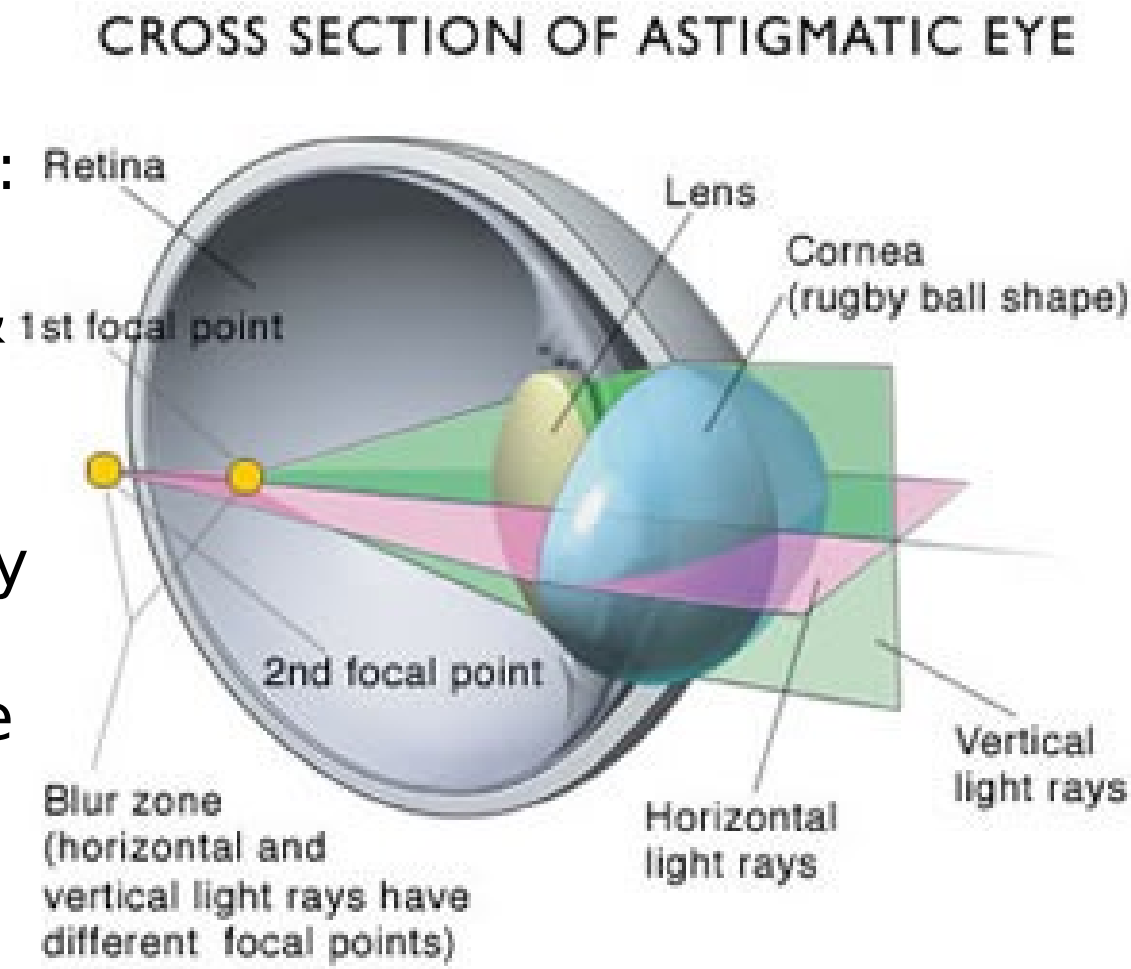
Definition:

- Lack of a focal point
- Curvature anomaly of the refractive media such that parallel incidence rays do not con-verge at a point but are drawn apart to form a line

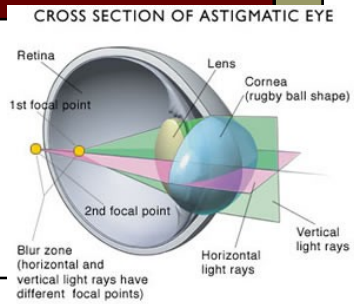


Astigmatism.....

- Definition.....:
  - Refractive media: (Ant. and Post. surface Cornea & Lens) are not spherical;
  - Refract differently along one meridian and along the meridian 90 degree to it: 2 focal points



Astigmatism……

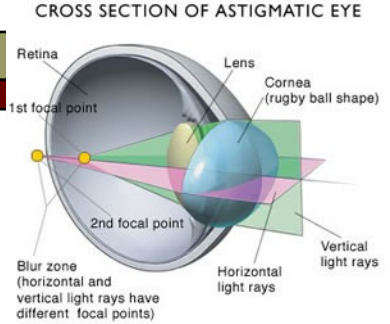


Epidemiology

- 42% of all humans $\geq \pm 0.5$D: 20% $\geq$ 1.00D

Classification

- External: (ant surface of the cornea) Internal: other components
- Rule
  - With the rule astigmatism: (most common)
    - meridian with the greatest refractive power is **Vertical (70 - 110 degrees)**
  - Against the rule astigmatism
    - meridian with the greatest refractive power is **Horizon-tal (160-20 degrees)**
  - Oblique astigmatism
- Regular/Irregular

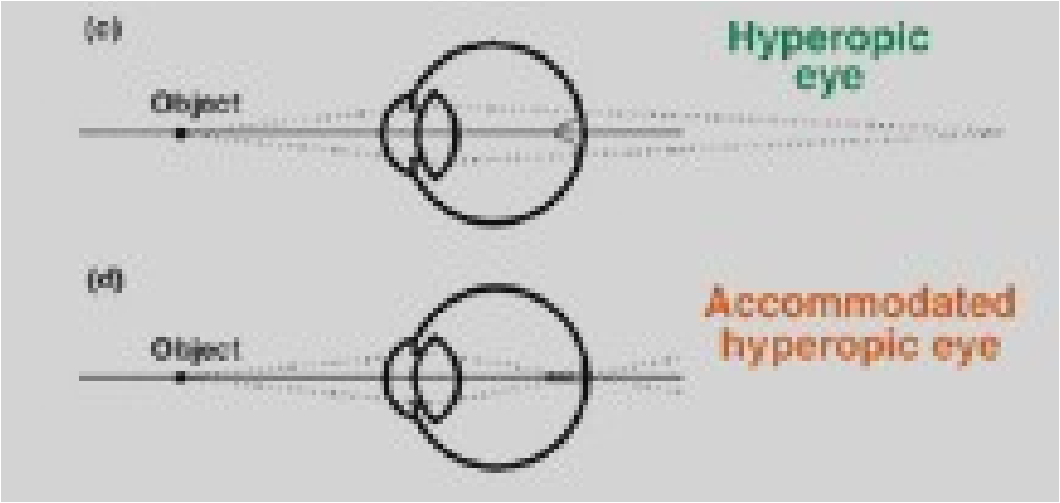


Astigmatism.....

- Symptoms
  - Distorted images
  - Asthenopic symp-toms
- Dignosis
  - Placidos disc
  - Corneal topogra-phy
  - Refraction: (Objec-tive, Subjective)
- Treatment
  - Cylindrical lenses (regular astigma-tism)
  - Contact lenses
    - Soft
    - Rigid
    - Toric

Presbyopia

- Definition
  - Physiological loss of accommodation with advanced age
  - Treatment – corrective spectacles (bifocal, multifocal)



Refractive Surgery

- Refractive Surgery
  - Photorefractive keratectomy (PRK)
  - **L**aser **A**ssisted **I**n-**S**itu **K**eratomileusis (LASIK)
  - LASEK
  - Refractive lens replacement – multifocal intraocular lenses, toric intraocular lenses
  - Intraocular Contact lens (**ICL)**

Refractive Surgery'.....

- LASIK is no different than Glasses or Contact lenses
  - It changes the way the light is focused as it enters the eye
  - By changing the shape of the cornea, the light is either converged or di-verged
  - Astigmatism is corrected by making the curvature of the cornea equal in all directions - making the surface of the eye round.

Thank you